WORKED SOLUTION

STEP II 2006, Question 3

Pure • Surds and Approximation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q3

The key idea

The number $5+\sqrt{24}$ has a "conjugate" $5-\sqrt{24}$, and the two multiply to 1. Adding equal powers of a number and its conjugate makes all the surds cancel, so the sum is an integer. The conjugate is small, so its powers are tiny, and the big power must sit just below an integer. Part (ii) is the same argument with 5 replaced by N.

Solution

Part (i)

Let $a = 5+\sqrt{24}$ and $b = 5-\sqrt{24}$. Then

$$a+b = 10, \qquad ab = 25-24 = 1,$$

so $b = 1/a$. The expression we want is therefore a^4+b^4. Using $a^2+b^2 = (a+b)^2-2ab$ twice:

$$\begin{aligned} a^2+b^2 &= 10^2-2 = 98,\\ a^4+b^4 &= (a^2+b^2)^2-2(ab)^2 = 98^2-2 = 9602. \end{aligned}$$

So $a^4+\dfrac{1}{a^4} = 9602$, an integer.

For the inequalities, recall that $\dfrac{1}{a} = b = 5-\sqrt{24}$.

- $b > 0.1 \iff \sqrt{24} < 4.9 \iff 24 < 24.01$, which is true. (Squaring is reversible here because both sides are positive.)
- $b < \frac{2}{19} \iff \sqrt{24} > 5-\frac{2}{19} = \frac{93}{19} \iff 24\cdot 19^2 > 93^2 \iff 8664 > 8649$, which is true.
- $\frac{2}{19} < 0.11 \iff 2 < 2.09$, which is true.

So $0.1 < \dfrac{1}{5+\sqrt{24}} < \dfrac{2}{19} < 0.11$.

Ask yourself How do these bounds pin down a^4 to four decimal places?

From the first result, $a^4 = 9602 - b^4$. Raising the positive inequalities $0.1 < b < 0.11$ to the fourth power,

$$0.0001 < b^4 < 0.11^4 = 0.00014641,$$

and therefore

$$9601.99985359 < a^4 < 9601.9999.$$

Every number in this interval lies between 9601.99985 and 9601.9999, so it rounds to

$$\left(5+\sqrt{24}\right)^4 = 9601.9999 \quad \text{(to four decimal places)}.$$

(The sharper bound $b < \frac{2}{19}$ gives $b^4 < \frac{16}{130321} \approx 0.000123$, which is not needed here but shows where $\frac{2}{19}$ comes from: see the Checks.)

Part (ii)

Let $A = N + \sqrt{N^2-1}$ and $B = N - \sqrt{N^2-1}$, both real since $N \geq 2$. As before,

$$A + B = 2N, \qquad AB = N^2 - (N^2-1) = 1.$$

Step 1: $A^k + B^k$ is an integer. Expand both powers by the binomial theorem. Terms with an even power of $\sqrt{N^2-1}$ are integers and appear identically in both; terms with an odd power appear with opposite signs and cancel. So $S_k = A^k + B^k$ is an integer for every positive integer k. (Alternatively, A and B are the roots of $x^2 - 2Nx + 1 = 0$, so $S_k = 2NS_{k-1} - S_{k-2}$ with $S_0 = 2$, $S_1 = 2N$, and induction gives integers.)

Step 2: B is small. We claim $0 < B < \dfrac{1}{2N - \frac{1}{2}}$. Since $B = 1/A$ and $A > 0$, this is equivalent to

$$A > 2N - \tfrac{1}{2} \iff \sqrt{N^2-1} > N - \tfrac{1}{2}.$$

Both sides are positive (as $N \geq 2$), so we may square:

$$N^2 - 1 > N^2 - N + \tfrac{1}{4} \iff N > \tfrac{5}{4},$$

which holds because $N \geq 2$. So $0 < B < \left(2N - \frac{1}{2}\right)^{-1}$, and raising to the kth power (everything positive),

$$0 < B^k < \left(2N - \tfrac{1}{2}\right)^{-k}.$$

Step 3: conclude. We have $A^k = S_k - B^k$, so A^k lies at distance exactly B^k from the integer S_k. The integer nearest to A^k is at least as close as S_k (in fact it is S_k, since $B^k < \frac{1}{2}$). Hence

$$\left|A^k - (\text{nearest integer})\right| \leq B^k < \left(2N - \tfrac{1}{2}\right)^{-k},$$

which is the required result.

Checks

- Numerically, $(5+\sqrt{24})^4 \approx 9601.999896$, inside our interval and rounding to 9601.9999.
- With $N=5$, part (ii) gives $(2N-\frac{1}{2})^{-1} = \frac{1}{9.5} = \frac{2}{19}$: exactly the bound from part (i). So part (i) is the case $N=5$, $k=4$ in disguise.
- For $N=2$, $k=1$: $2+\sqrt{3} \approx 3.732$ is about 0.268 from 4, and $(3.5)^{-1} \approx 0.286$, so the bound holds. A computer checked N up to 1000 and k up to 24.

Common slips

- Squaring $\sqrt{N^2-1} > N-\frac{1}{2}$ without noting both sides are positive.
- Writing $a^4 \approx 9602.0000$: the value is *below* 9602 by between 0.0001 and 0.00015, and that is enough to change the fourth decimal place.
- Forgetting that the case $N=1$ is excluded for a reason: then $B=1$ and the argument collapses.

Beyond STEP

Numbers like $5+\sqrt{24}$ are units in the ring $\mathbb{Z}[\sqrt{6}]$: they have inverses $5-\sqrt{24}$ of the same form, and their powers give all the solutions of the Pell equation $x^2-24y^2=1$. Powers of such units get ever closer to integers, and this is one route to the theory of Pisot numbers, whose powers have the same near-integer behaviour. Davenport's *The Higher Arithmetic* treats Pell's equation and continued fractions beautifully.

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